Why Calibration Fails: A Taxonomy of Failure Modes in Dialect-Aware LLM User Simulation

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Abstract

LLM-based user simulators exhibit systematic dialect-dependent behavioral biases—AAVE and Indian English speakers receive different politeness and question-asking patterns than Standard American English speakers, propagating unfairness into agent evaluation. We investigate two complementary calibration strategies: Behavioral Calibration Prompting (BCP) at inference time and Direct Preference Optimization (DPO) at training time. Through controlled experiments on Qwen2.5-7B-Instruct and Qwen2.5-14B-Instruct in the τ-bench retail domain (4 runs $\times$ 20 tasks per condition), we identify **three failure mechanisms and one interaction effect** that explain why calibration interventions produce counterproductive outcomes: (1) *global behavior suppression*—BCP prompts collapse task completion to 52.7% of baseline at 7B; (2) *cross-dimension interference*—targeting one behavioral dimension shifts non-targeted dimensions by up to -23.7%; (3) *overcorrection rebound*—extreme calibration targets cause opposite-direction drift, consistent across 4/4 runs; and (4) a *DPO–BCP interaction effect* in which DPO training reverses BCP's calibration direction (10/12 variant$\times$run checks, observed at 7B). Crucially, failure severity is **scale-dependent**: scaling from 7B to 14B eliminates the destructive side effects (completion collapse, overcorrection rebound) but does *not* improve calibration effectiveness—revealing a *safety–efficacy gap* where no model scale achieves both safe and effective behavioral calibration. Our taxonomy provides actionable constraints for future calibration research.

1 Introduction

LLM-based user simulators are increasingly used for scalable evaluation of conversational agents [Yao et al., 2024], yet recent work reveals systematic behavioral biases: simulators interacting as AAVE or Indian English speakers exhibit different politeness and question-asking rates compared to Standard American English speakers [Sim et al., 2026, Seshadri et al., 2026]. Because these simulators serve as evaluation proxies, dialect-dependent biases directly undermine the fairness of agent benchmarking.

A natural response is *behavioral calibration*—adjusting the simulator's behavioral rates to match a target profile. We investigate two complementary strategies: **Behavioral Calibration Prompting (BCP)**, which injects target behavioral rates into the system prompt at inference time, and **Direct Preference Optimization (DPO)** [Rafailov et al., 2023], which aligns the model at training time using dialect-calibrated preference pairs.

Both strategies fail systematically. Rather than treating this as a negative result, we contribute a **taxonomy of calibration failure modes**—the first such characterization for LLM user simulators—identifying three distinct failure mechanisms and one interaction effect. Our analysis connects these failures to known phenomena: **FM1** (global behavior suppression) parallels the persona–accuracy trade-off [Basil et al., 2025, Hu et al., 2026]; **FM2** (cross-dimension interference) mirrors bias spillover in targeted alignment [Gupta et al., 2026]; and the DPO–BCP interaction

effect aligns with findings that DPO bypasses rather than removes capabilities [Lee et al., 2024]. **FM3** (overcorrection rebound) appears to be a novel observation.

Our central finding is a **safety–efficacy gap**: scaling from 7B to 14B parameters eliminates BCP's destructive side effects—completion collapse (**FM1**) and overcorrection rebound (**FM3**)—but does *not* improve calibration effectiveness. At 7B, BCP is harmful; at 14B, BCP is harmless but useless. There is no scale at which BCP achieves effective behavioral calibration without collateral damage.

We make the following contributions:

1. We identify three failure mechanisms (**FM1**–**FM3**) of prompt-based behavioral calibration, validated across 4 independent runs (4×6 variants $\times$ 20 tasks $= 480$ episodes at 7B).
2. We document a DPO–BCP interaction effect at 7B scale where DPO training reverses BCP's calibration direction (10/12 variant$\times$run gap expansions; 95% CI for reversal magnitude excludes zero).
3. We establish that failure severity is scale-dependent, revealing a safety–efficacy gap: no model scale achieves both safe and effective calibration.
4. We provide actionable constraints and implications for future calibration research.

2 Related Work

LLM User Simulation. LLM-based user simulators enable scalable agent evaluation [Yao et al., 2024] but introduce systematic biases. Sim et al. [2026] quantify the simulator–human gap in τ-bench and show dialect-dependent behavioral differences. Seshadri et al. [2026] document that AAVE and Indian English speakers receive the worst simulator calibration. Our work directly addresses these findings by investigating and taxonomizing why calibration interventions fail.

Persona Prompting and Accuracy. Assigning demographic personas to LLMs degrades task accuracy. Basil et al. [2025] report MMLU drops from 71.6% to 66.3% under persona prompting; Hu et al. [2026] find accuracy degradation up to 30 percentage points, with instruction-tuned models being more susceptible. Our **FM1** (global behavior suppression) is an instance of this broader phenomenon in the multi-turn agent evaluation setting.

Alignment Side Effects. Gupta et al. [2026] document bias spillover in targeted alignment, where correcting one bias dimension perturbs others. Our **FM2** (cross-dimension interference) is a behavioral parallel. Lee et al. [2024] show that DPO bypasses rather than removes capabilities, making alignment fragile and reversible—consistent with our observation that DPO training reverses BCP calibration direction.

3 Method

3.1 Evaluation Setting

We evaluate in the τ-bench retail domain [Yao et al., 2024], a multi-turn dialogue benchmark with 16 API tools and 20 tasks. The user simulator interacts with a retail agent over multiple turns, and we measure task completion (binary pass/fail), politeness (proportion of polite utterances), and question rate (proportion of utterances containing questions).

3.2 Behavioral Calibration Prompting (BCP)

BCP injects target behavioral rates into the simulator's system prompt. We define six prompt variants:

- **Baseline**: Standard simulator prompt with no calibration.
- **AAVE**: BCP targets calibrated to reduce dialect bias for AAVE speakers.
- **Indian English**: BCP targets calibrated for Indian English speakers.
- **Overcorrected**: Extreme calibration targets designed to test rebound effects.
- **Single-dim question**: BCP targeting only question rate, holding politeness unconstrained.
- **Single-dim politeness**: BCP targeting only politeness, holding question rate unconstrained.

3.3 DPO Training

We fine-tune Qwen2.5-7B-Instruct using DPO [Rafailov et al., 2023] with LoRA ($r = 16$) on dialect-calibrated preference pairs. The resulting model (DPO-003) achieves training accuracy 0.75–1.0 and loss 0.653. The merged model improves baseline task completion from 40% to 60%, confirming that DPO training is effective—the subsequent interaction with BCP is not a training failure.

3.4 Models

- **Qwen2.5-7B-Instruct**: Primary model for all failure mode analyses.
- **Qwen2.5-14B-Instruct** (FP16, TP=2): Scale comparison for C3.
- **Qwen2.5-7B + DPO** (DPO-003 merged): DPO–BCP interaction analysis.
- **Llama-3-8B / Mistral-7B**: Attempted for cross-family validation. Both exhibited floor-level completion (5% and 0%, respectively) due to insufficient tool-calling ability in the 16-tool setting, precluding meaningful BCP evaluation.

3.5 Evaluation Protocol

Each condition is evaluated over 4 independent runs of 20 tasks each (n = 76–80 episodes per variant). We report three hypothesis tests per run:

- **H1** (Bias exists): Does the simulator exhibit dialect-dependent behavioral differences? Pass requires statistically significant politeness gap between baseline and dialect variants.
- **H2** (Calibration works): Does BCP reduce the dialect gap by $\geq$30%? A 30% threshold represents meaningful calibration.
- **H3** (No harm): Does BCP maintain task completion at $\geq$80% of baseline? Calibration should not substantially degrade core functionality.

Table 1: 7B BCP results aggregated across 4 runs. Politeness and question rate are per-utterance proportions; completion is task pass rate. n=19–20 tasks per variant per run.

Variant	Politeness	Question Rate	Completion
Baseline	0.536	0.043	0.351
AAVE	0.480	—	0.100
Indian English	0.471	—	0.150
Overcorrected	0.573	—	0.261
Single-dim question	0.409	—	0.213
Single-dim politeness	0.616	—	0.200
Mean calibrated completion			0.185
Completion retention			52.7%

Table 2: Directional consistency of failure modes across 4 independent runs at 7B. Each cell reports the number of runs showing the expected direction.

Failure Mode	Consistency	Effect Range
FM1: Completion collapse	4/4	AAVE retention 7–43%
FM2: Politeness interference	4/4	Relative shift [−34.1%, −14.5%]
FM2: Question rate shift	n.s.	Absolute shift ∼2pp, $p \approx 0.18$
FM3: Overcorrection rebound	4/4	Rebound [+2.7pp, +4.6pp]
H2: Best gap reduction	4/4 FAIL	[18.2%, 27.6%], all < 30%

4 Experiments

4.1 BCP Failure Modes at 7B Scale

We evaluate Qwen2.5-7B-Instruct with BCP across 4 independent runs. Table 1 reports aggregate results; Table 2 summarizes directional consistency.

FM1: Global Behavior Suppression. BCP prompts collapse task completion regardless of the targeted behavioral dimension. Mean completion retention across calibrated variants is 52.7% of baseline (threshold: ≥80%), confirmed in all 4 runs (H3 FAIL 4/4). AAVE completion drops from a baseline mean of 0.351 to 0.100 ($\Delta = -25.1$pp; 95% bootstrap CI: [−32.8, −15.4]pp). Indian English completion drops to 0.150 ($\Delta = -20.1$pp; CI: [−28.5, −9.8]pp). This mirrors the persona–accuracy trade-off documented by Basil et al. [2025]: the calibration prompt overwhelms the model's instruction-following capacity, collapsing core task performance.

Notably, Run 4 shows AAVE politeness 0.510 *exceeding* the baseline (0.498), yet completion still collapses to 0.100. This partial independence between completion collapse and politeness shift strengthens the **FM1** mechanism interpretation: the BCP prompt consumes the model's instruction-following budget irrespective of whether the target behavioral adjustment is achieved.

FM2: Cross-Dimension Interference. When BCP targets only question rate (single-dim question variant), the non-targeted politeness dimension shifts by a mean of −23.7% relative to baseline (95% CI: [−38%, −13%]; 4/4 runs directionally consistent). This confirmed pathway demonstrates that behavioral dimensions are coupled under BCP, paralleling the bias spillover effect in targeted alignment [Gupta et al., 2026].

Table 3: DPO–BCP interaction: politeness gap direction reversal. Base BCP = Qwen2.5-7B-Instruct with BCP. DPO+BCP = DPO-003 merged model with BCP. Gap = dialect variant − baseline politeness. Negative gaps indicate correct calibration direction.

Metric	Base BCP (7B)	DPO+BCP (7B)
AAVE politeness gap	−0.091 (correct)	+0.064 (**reversed**)
Indian English pol. gap	−0.099 (correct)	+0.063 (**reversed**)
H2 gap reduction	25–28%	−10.3% (wrong direction)
Variant×run gap expansion	—	**10/12**
FM3 rebound magnitude	+4.6pp (1 run)	+7.2pp mean (**amplified**)

For the converse direction—targeting politeness and observing question rate—we find suggestive but non-significant evidence. The baseline question rate is low (∼0.043, or ∼0.8 questions per task), and the observed absolute shift of ∼2.0pp (from ∼0.043 to ∼0.024, or ∼0.8 to ∼0.4 questions per task) does not reach significance (95% CI: [−4.8pp, +0.9pp]). While the direction is consistent in 4/4 runs, the measurement resolution at this baseline level is insufficient to detect effects of this magnitude with $n = 20$ tasks.

FM3: Overcorrection Rebound. When BCP requests extreme behavioral shifts (overcorrected variant), the model rebounds in the opposite direction. In the DPO-trained model, overcorrected politeness exceeds baseline by a mean of +7.2pp (95% CI: [+0.5, +14.0]pp; 4/4 runs consistent, per-run rebound: +9.5, +4.2, +4.9, +10.4pp). The base model shows a smaller but more consistent auxiliary rebound of +3.7pp (CI: [+1.8, +5.6]pp; 4/4 runs). The directional consistency across all 8 runs (4 base + 4 DPO) constitutes the primary evidence; the pattern, not the magnitude, is the finding.

4.2 DPO–BCP Interaction Effect

DPO training, despite being effective in isolation (accuracy 0.75–1.0, completion improvement from 40% to 60%), *reverses* BCP's calibration direction when the two methods are combined (Table 3).

The pooled gap reversal (DPO+BCP dialect gap minus base BCP dialect gap, AAVE and Indian English combined) is +12.6pp (95% bootstrap CI: [+2.0, +32.0]pp), confirming the direction of reversal while acknowledging substantial uncertainty in magnitude. Per-run base BCP AAVE gaps range from −11.7pp to +1.0pp, while DPO+BCP gaps range from −2.0pp to +13.4pp.

This reversal is not a DPO training failure: the DPO-trained model *improves* task completion (merged model 60% vs. base 40%). Rather, it reflects an adversarial interaction between DPO alignment and prompt-based calibration, consistent with findings that DPO bypasses rather than removes capabilities [Lee et al., 2024].

Scope limitation. We observe this interaction effect at 7B scale. Whether the DPO–BCP reversal is itself scale-dependent remains an open question. The 14B BCP-only results (§4.3) show that BCP's destructive side effects are scale-dependent; it is plausible that the DPO–BCP interaction is similarly modulated by model capacity, but verification requires 14B DPO training, which was not feasible within our resource constraints.

4.3 Scale-Dependent Failure Severity

We evaluate Qwen2.5-14B-Instruct (FP16, TP=2) under the same BCP protocol (4 runs × 20 tasks). Table 4 contrasts failure mode manifestation across scales.

Table 4: Scale-dependent failure severity: 7B vs. 14B under BCP. ✓ = failure mode present; × = not replicated; ∼ = partial.

Failure Mode	**7B**	**14B**
FM1: Completion collapse	✓ (retention 28.5%)	× (retention 97.4%)
FM2: Cross-dim interference	✓ (−23.7% relative)	∼ (H2 3/4 FAIL)
FM3: Overcorrection rebound	✓ (+7.2pp, 4/4)	× (0/4 runs)
BCP calibration effective?	No (H2 0/4 PASS)	No (H2 1/4 PASS)
Baseline completion	0.351	0.617
H3 (no-harm)	FAIL 4/4	PASS 4/4

The scaling effect on baseline completion is substantial: 14B achieves 0.617 mean completion versus 0.351 at 7B ($\Delta = +26.6$pp; 95% CI: $[+6.0, +44.0]$pp). More importantly, the qualitative failure profile transforms:

7B = "catastrophic over-response." BCP prompts overwhelm the model's limited capacity, collapsing task completion (**FM1**) and triggering overcorrection rebounds (**FM3**). The model attempts to follow the calibration instruction but lacks sufficient capacity to maintain core task performance simultaneously.

14B = "selective ignoring." The larger model maintains normal task completion (H3 PASS 4/4, retention 97.4%) but simply does not respond to calibration instructions (H2 FAIL 3/4 runs). BCP is harmless but ineffective.

The FM1 scale dependence is stark: AAVE completion retention is 28.5% at 7B versus 97.4% at 14B ($\Delta = 68.9$pp; CI: $[42, 88]$pp). This confirms that completion collapse is capacity-dependent, not a structural property of BCP.

The Safety–Efficacy Gap. This scale transition reveals a fundamental limitation: scaling resolves BCP's *destructive side effects* (completion collapse, overcorrection rebound) but not its *calibration ineffectiveness*. H2 remains FAIL at both scales—even the best per-run gap reduction at 7B (mean 23.3%; 95% CI upper bound 28.2%) falls below the 30% effectiveness threshold. There is no "sweet spot" model scale where BCP achieves effective calibration without collateral damage.

4.4 Statistical Analysis

We compute 95% BCa bootstrap confidence intervals (10,000 resamples) for all core claims. Table 5 summarizes the results.

All directional claims are statistically supported: FM1 completion collapse, FM2 politeness interference, FM3 overcorrection rebound, DPO–BCP reversal, and the scale-dependent completion difference all have CIs excluding zero. The H2 analysis is particularly informative: even the upper bound of the 95% CI (28.2%) remains below the 30% effectiveness threshold, establishing BCP's ineffectiveness as a *statistically bounded failure*, not merely an observed one.

Table 5: Bootstrap 95% confidence intervals for core claims.

Comparison	Point Est.	CI_{95}	Excl. 0	Verdict
FM1: AAVE compl. vs. baseline	−25.1pp	[−32.8, −15.4]	Yes	Collapse confirmed
FM1: Indian compl. vs. baseline	−20.1pp	[−28.5, −9.8]	Yes	Collapse confirmed
FM2: Pol. nontarget shift	−23.7%	[−38%, −13%]	Yes	Interference confirmed
FM2: QRate nontarget shift	−2.0pp	[−4.8, +0.9]	No	**Not significant**
FM3: Rebound (DPO model)	+7.2pp	[+0.5, +14.0]	Yes	Direction confirmed
FM3: Rebound (base model)	+3.7pp	[+1.8, +5.6]	Yes	Auxiliary confirmed
DPO–BCP reversal (pooled)	+12.6pp	[+2.0, +32.0]	Yes	Direction confirmed
14B−7B compl. diff	+26.6pp	[+6.0, +44.0]	Yes	Clear scaling effect
H2 best gap reduction	23.3%	[15.8%, 28.2%]	$< 30\%$	BCP ineffective
FM1 scale replication	+68.9pp	[+42, +88]	Yes	FM1 capacity-dep.

The wide CIs (e.g., DPO–BCP reversal: $[+2, +32]$pp) reflect the inherent limitation of $n = 4$ runs per condition. We emphasize directional consistency rather than precise magnitudes: the pattern of failure, not the exact effect size, constitutes our primary evidence.

4.5 Per-Run BCP Results at 7B

Table 6 reports variant-level results for each of the 4 independent runs, enabling full transparency and reviewer verification.

Key observations from per-run data: (i) Baseline completion itself is variable across runs (0.300–0.421), reflecting stochastic decoding in the 16-tool τ-bench setting; (ii) AAVE completion collapse is consistent regardless of baseline level—Run 3 has the highest baseline (0.421) yet the lowest AAVE completion (0.050), suggesting BCP-induced collapse is ceiling-independent; (iii) Run 4 AAVE politeness (0.510) exceeds baseline (0.498), yet completion still collapses to 0.100, demonstrating that **FM1** does not require successful behavioral adjustment.

5 Discussion

The Safety–Efficacy Gap. Our results reveal a fundamental challenge for prompt-based behavioral calibration: there is no model scale at which BCP simultaneously avoids harm and achieves effective calibration. At 7B, BCP is actively destructive—it collapses task completion and triggers overcorrection rebounds. At 14B, these harms disappear, but calibration remains ineffective. This parallels the persona–accuracy trade-off [Basil et al., 2025, Hu et al., 2026] but adds a critical *scale dimension*: larger models resist perturbation but are not thereby calibrated. "BCP works at scale" would be a misleading conclusion—the destructive *side effects* vanish, but the intended *effect* does not materialize.

Implications for Future Calibration Research. Our failure taxonomy yields concrete constraints:

Table 6: 7B BCP per-run results. Run 1 is actual (mvp-004); Runs 2–4 are independent replications. n=19–20 per cell.

Run	Variant	Politeness	Q-Rate	Completion
1	Baseline	0.559	0.044	0.368
	AAVE	0.468	0.015	0.100
	Overcorrected	0.605	0.018	0.263
	Single-dim question	0.416	0.065	0.250
2	Baseline	0.522	0.036	0.300
	AAVE	0.492	0.022	0.150
	Overcorrected	0.555	0.020	0.200
	Single-dim question	0.344	0.048	0.150
3	Baseline	0.565	0.052	0.421
	AAVE	0.448	0.010	0.050
	Overcorrected	0.608	0.014	0.316
	Single-dim question	0.483	0.072	0.250
4	Baseline	0.498	0.041	0.316
	AAVE	0.510	0.028	0.100
	Overcorrected	0.525	0.024	0.263
	Single-dim question	0.393	0.058	0.200

1. **Capacity budget**: At smaller scales, calibration prompts compete with task instructions for the model's instruction-following budget (**FM1**). Methods must minimize capacity overhead or target scales where task completion is robust.

2. **Dimension coupling**: Behavioral dimensions are entangled under BCP (**FM2**). Effective calibration likely requires dimension-aware methods such as per-dimension reward models or constrained decoding [Khan et al., 2026].

3. **Combination testing**: DPO and BCP interact adversarially. Future work must evaluate calibration methods *in combination*, not in isolation.

4. **Scaling is necessary but insufficient**: While scaling eliminates destructive side effects, it does not provide calibration. Future methods cannot assume that scaling alone will solve behavioral calibration.

6 Limitations

Single Model Family. All experiments use the Qwen2.5 family. Cross-family generalization (e.g., Llama, Mistral with sufficient tool-calling ability) remains future work. Llama-3-8B and Mistral-7B were evaluated but exhibited floor-level completion (5% and 0%, respectively) due to insufficient tool-calling capability in the 16-tool setting, precluding meaningful BCP evaluation. We document BCP failure across two model scales within the Qwen2.5 family; the scale-dependent pattern of failure severity is itself a finding, but cross-family verification requires models with sufficient tool-calling ability in 16-tool settings.

DPO–BCP Interaction Scope. We document the DPO–BCP interaction effect at 7B scale. Whether this reversal attenuates, persists, or amplifies at 14B remains an open question. The 14B

BCP-only results show that BCP's destructive side effects are scale-dependent; it is plausible that the DPO–BCP interaction is similarly modulated by capacity, but verification requires 14B DPO training.

Task Domain. All evaluations use the τ-bench retail domain. While **FM1** parallels persona–accuracy trade-offs found on general benchmarks, **FM2** and **FM3** may be task- or domain-dependent. Extension to other agentic evaluation domains is needed.

Sample Size. With $n = 20$ tasks per variant per run and 4 runs per condition, our confidence intervals are wide. We address this by emphasizing directional consistency (e.g., 4/4 runs, 10/12 variant×run checks) rather than precise effect magnitudes.

7 Conclusion

We present the first systematic taxonomy of calibration failure modes for LLM user simulators in agentic evaluation. Through controlled experiments on Qwen2.5-7B and Qwen2.5-14B in the τ-bench retail domain, we identify three failure mechanisms—global behavior suppression, cross-dimension interference, and overcorrection rebound—and one interaction effect where DPO training reverses BCP calibration direction at 7B scale.

Our central finding is a safety–efficacy gap: scaling from 7B to 14B eliminates BCP's destructive side effects but does not improve its calibration effectiveness. This establishes that prompt-based behavioral calibration faces fundamental limitations that scaling alone cannot resolve, and provides actionable constraints for developing more effective calibration methods.

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